

System Architecture Overview

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This document describes the production deployment topology, capacity planning and disaster recovery architecture for all customer-facing services.

1. Deployment Topology

1.1 Production is deployed on the public cloud across 3 availability zones in a single region.

1.2 Every service runs at least 3 replicas, spread evenly across availability zones.

1.3 The API gateway is the single entry point for external traffic and terminates TLS.

1.4 Internal services communicate over gRPC; asynchronous messages flow through Kafka.

1.5 The data platform uses ClickHouse for analytics and Elasticsearch for full-text search.

1.6 MySQL is used for transactional data that requires strong consistency.

2. Capacity Planning

2.1 The system is designed for a peak load of 50,000 QPS at the gateway layer.

2.2 A single service instance must not exceed 80% CPU or 75% memory utilisation.

2.3 Auto-scaling is triggered when CPU utilisation exceeds 70% for 5 consecutive minutes.

2.4 Database connection pools are sized at 100 connections per service instance.

2.5 Cache clusters use Redis with 6 nodes in a 3-master 3-replica layout.

3. Disaster Recovery

3.1 Recovery Point Objective (RPO) is 15 minutes; Recovery Time Objective (RTO) is 4 hours.

3.2 Full database backups run daily; incremental backups run every hour.

3.3 Backup files are stored in object storage in a different region.

3.4 Disaster recovery drills are conducted every quarter; results are reviewed by the SRE team.

4. Middleware Selection

4.1 Kafka is used for event streaming; consumer groups must enable exactly-once semantics where supported.

4.2 Redis is used for hot caches with a maximum key TTL of 24 hours.

4.3 Elasticsearch indices follow a daily rolling pattern with 30 days retention.

4.4 ClickHouse is used only for append-only analytical workloads.